

Leibniz Integral Rule

Senan Sekhon

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0 The Leibniz Integral Rule

Theorem 0.1 (Leibniz Integral Rule – Riemann Integral). Suppose $I \subseteq \mathbb{R}$ is an open interval and $f : [a, b] \times I \rightarrow \mathbb{R}$, $(x, t) \mapsto f(x, t)$ such that f and $\frac{\partial f}{\partial t}$ are continuous on $[a, b] \times I$. Define $S : I \rightarrow \mathbb{R}$ as follows:

$$S(t) = \int_a^b f(x, t) dx$$

Then S is differentiable on I and:

$$S'(t) = \int_a^b \frac{\partial f}{\partial t}(x, t) dx$$

Proof. Suppose $t_0 \in I$ and $\varepsilon > 0$. Since I is an open interval, there exists $\delta_0 > 0$ such that $[t_0 - \delta_0, t_0 + \delta_0] \subseteq I$. Since $\frac{\partial f}{\partial t}$ is continuous on $[a, b] \times I$, by the **Heine-Cantor theorem**, it is uniformly continuous on $[a, b] \times [t_0 - \delta_0, t_0 + \delta_0]$. Thus there exists $0 < \delta < \delta_0$ such that:

$$\left| \frac{\partial f}{\partial t}(x, s) - \frac{\partial f}{\partial t}(x, t_0) \right| < \frac{\varepsilon}{b-a} \text{ for all } x \in [a, b] \text{ and all } s \in I, |s - t_0| < \delta$$

Now suppose $t \in I$, $|t - t_0| < \delta$ and $x \in [a, b]$. By **Lagrange's mean value theorem**, there exists ξ (possibly depending on x) between t_0 and t such that $\frac{f(x, t) - f(x, t_0)}{t - t_0} = \frac{\partial f}{\partial t}(x, \xi)$. This yields:

$$\left| \frac{f(x, t) - f(x, t_0)}{t - t_0} - \frac{\partial f}{\partial t}(x, t_0) \right| = \left| \frac{\partial f}{\partial t}(x, \xi) - \frac{\partial f}{\partial t}(x, t_0) \right| < \frac{\varepsilon}{b-a}$$

Note that the above inequality holds for all $x \in [a, b]$, even though ξ may depend on x . We also have:

$$\begin{aligned} \frac{S(t) - S(t_0)}{t - t_0} - \int_a^b \frac{\partial f}{\partial t}(x, t_0) dx &= \frac{1}{t - t_0} \left(\int_a^b f(x, t) dx - \int_a^b f(x, t_0) dx \right) - \int_a^b \frac{\partial f}{\partial t}(x, t_0) dx \\ &= \int_a^b \left(\frac{f(x, t) - f(x, t_0)}{t - t_0} - \frac{\partial f}{\partial t}(x, t_0) \right) dx \end{aligned}$$

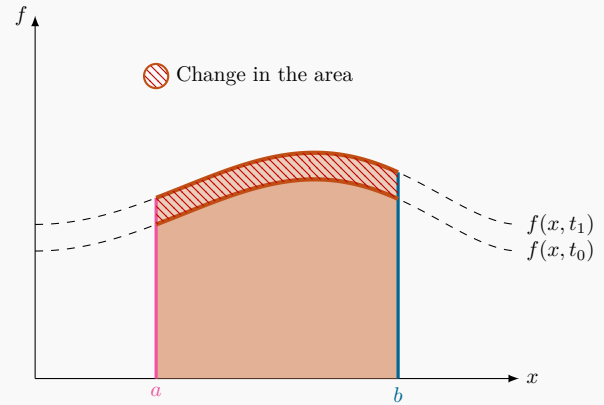
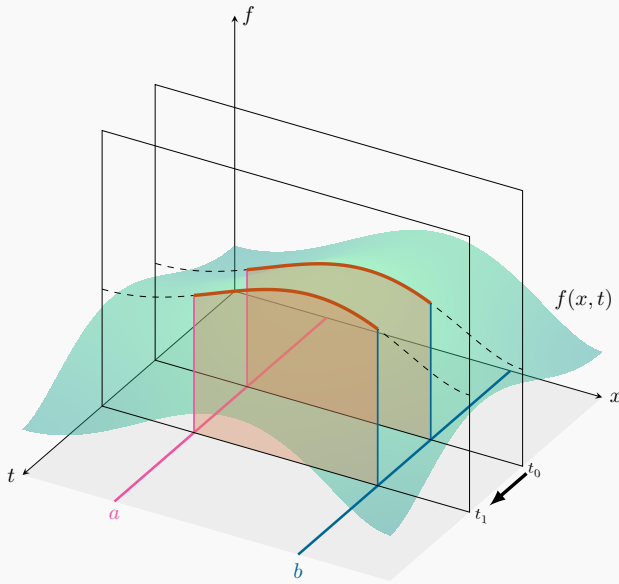
This yields:

$$\left| \frac{S(t) - S(t_0)}{t - t_0} - \int_a^b \frac{\partial f}{\partial t}(x, t_0) dx \right| \leq \int_a^b \left| \frac{f(x, t) - f(x, t_0)}{t - t_0} - \frac{\partial f}{\partial t}(x, t_0) \right| dx < \int_a^b \frac{\varepsilon}{b-a} dx = (b-a) \frac{\varepsilon}{b-a} = \varepsilon$$

Thus S is differentiable at t_0 and $S'(t_0) = \int_a^b \frac{\partial f}{\partial t}(x, t_0) dx$. 🤖

The Leibniz Integral Rule

$$\frac{d}{dt} \left(\int_a^b f(x, t) dx \right) = \int_a^b \frac{\partial f}{\partial t}(x, t) dx$$



As t moves from t_0 to t_1 , the shaded area changes from $\int_a^b f(x, t_0) dx$ to $\int_a^b f(x, t_1) dx$.

Dividing this by $t_1 - t_0$ and taking the limit as $t_1 - t_0$ becomes small, we get $\int_a^b \frac{\partial f}{\partial t}(x, t) dx$.

Theorem 0.2 (Leibniz Integral Rule – Improper Riemann Integral). Suppose $I \subseteq \mathbb{R}$ is an open interval and $f : [a, \infty) \times I \rightarrow \mathbb{R}$, $(x, t) \mapsto f(x, t)$ such that:

- f is continuous on $[a, \infty) \times I$ improperly Riemann integrable with respect to x on $[a, \infty)$ for each $t \in I$.
- $\frac{\partial f}{\partial t}$ is continuous on $[a, \infty) \times I$ and improperly Riemann integrable with respect to x on $[a, \infty)$, uniformly in $t \in I$. In other words, for any $\varepsilon > 0$, there exists $M > a$ (independent of t) such that for all $y > M$ and all $t \in I$, we have $|\int_a^y f(x, t) dx| < \varepsilon$.

Define $S : I \rightarrow \mathbb{R}$ as follows:

$$S(t) = \int_a^\infty f(x, t) dx$$

Then S is differentiable on I and:

$$S'(t) = \int_a^\infty \frac{\partial f}{\partial t}(x, t) dx$$

Proof. Suppose $(b_n)_{n=1}^\infty$ is a sequence in (a, ∞) and $\lim_{n \rightarrow \infty} b_n = \infty$. For each $n \in \mathbb{N}$, define $S_n : I \rightarrow \mathbb{R}$ as follows:

$$S_n(t) = \int_a^{b_n} f(x, t) dx$$

By **Theorem 0.1**, for each $n \in \mathbb{N}$, we have $S'_n(t) = \int_a^{b_n} \frac{\partial f}{\partial t}(x, t) dx$. Since $\frac{\partial f}{\partial t}(x, t)$ is improperly Riemann integrable with respect to x on $[a, \infty)$, uniformly in t , we have:

$$\lim_{n \rightarrow \infty} S'_n(t) = \int_a^\infty \frac{\partial f}{\partial t}(x, t) dx$$

FINISH THE PROOF!!!



The Leibniz integral rule can be applied to known integrals to derive related formulas:

Example.

$$\begin{aligned}
\int_0^\infty \frac{1}{x^2 + a^2} dx &= \frac{\pi}{2a} & \xrightarrow{\frac{d}{da}} & \int_0^\infty \frac{1}{(x^2 + a^2)^2} dx = \frac{\pi}{4a^3} \\
& & \xrightarrow{\frac{d}{da}} & \int_0^\infty \frac{1}{(x^2 + a^2)^3} dx = \frac{3\pi}{16a^5} \\
& & \vdots & \\
& & \xrightarrow{\frac{d}{da}} & \int_0^\infty \frac{1}{(x^2 + a^2)^n} dx = \binom{2n-2}{n-1} \frac{\pi}{(2a)^{2n-1}}
\end{aligned}$$

Example.

$$\begin{aligned}
\int_0^\infty e^{-ax} dx &= \frac{1}{a} & \xrightarrow{\frac{d}{da}} & \int_0^\infty x e^{-ax} dx = \frac{1}{a^2} \\
& & \xrightarrow{\frac{d}{da}} & \int_0^\infty x^2 e^{-ax} dx = \frac{2}{a^3} \\
& & \vdots & \\
& & \xrightarrow{\frac{d}{da}} & \int_0^\infty x^n e^{-ax} dx = \frac{n!}{a^{n+1}}
\end{aligned}$$

The Leibniz integral rule can also be used to solve certain integrals, by differentiating them first:

Example.

$$S(t) = \int_0^1 \frac{x^t - 1}{\log(x)} dx \quad (t > -1)$$

Differentiating under the integral, we get:

$$S'(t) = \int_0^1 x^t dx = \frac{1}{t+1}$$

Integrating yields:

$$S(t) = \log(t+1) + C$$

Where C is a constant. Note that the integrand is zero when $t = 0$, so $S(0) = 0$ and so $C = 0$. Thus:

$$\int_0^1 \frac{x^t - 1}{\log(x)} dx = \log(t+1)$$

Example (Dirichlet integral).

$$S(t) = \int_0^\infty e^{-tx} \frac{\sin(x)}{x} dx \quad (t > 0)$$

Differentiating under the integral, we get:

$$S'(t) = - \int_0^\infty e^{-tx} \sin(x) dx = - \frac{1}{t^2 + 1}$$

Integrating yields:

$$S(t) = -\arctan(t) + C$$

Where C is a constant. Note that the integrand tends to zero as $t \rightarrow \infty$, so $\lim_{t \rightarrow \infty} S(t) = 0$ and so $C = \frac{\pi}{2}$. Thus:

$$\int_0^\infty e^{-tx} \frac{\sin(x)}{x} dx = \frac{\pi}{2} - \arctan(t)$$

Taking the limit as $t \rightarrow 0^+$ yields:

$$\int_0^\infty \frac{\sin(x)}{x} dx = \frac{\pi}{2}$$

The following integral was solved in 1844 by [Joseph Serret](#) (1819 – 1885).

Example (Serret's Integral). Consider the following integral:

$$I = \int_0^1 \frac{\log(1+x)}{1+x^2} dx$$

To evaluate this, we first define $S(t)$ as follows:

$$S(t) = \int_0^1 \frac{\log(1+tx)}{1+x^2} dx$$

Differentiating under the integral, we get:

$$S'(t) = \int_0^1 \frac{x}{(1+tx)(1+x^2)} dx = -\frac{\log(1+t)}{1+t^2} + \frac{\pi t}{4(1+t^2)} + \frac{\log(2)}{2(1+t^2)}$$

Note that $S(0) = 0$ and $S(1) = I$. Thus, we have:

$$\begin{aligned} I = S(1) - S(0) &= \int_0^1 S'(t) dt \\ &= -\int_0^1 \frac{\log(1+t)}{1+t^2} dt + \frac{\pi}{4} \int_0^1 \frac{t}{1+t^2} dt + \frac{\log(2)}{2} \int_0^1 \frac{1}{1+t^2} dt \\ &= -\int_0^1 \frac{\log(1+t)}{1+t^2} dt + \frac{\pi}{4} \left(\frac{\log(2)}{2} \right) + \frac{\log(2)}{2} \left(\frac{\pi}{4} \right) \\ &= -I + \frac{\pi}{4} \log(2) \end{aligned}$$

Rearranging yields $2I = \frac{\pi}{4} \log(2)$. Thus:

$$\int_0^1 \frac{\log(1+x)}{1+x^2} dx = \frac{\pi}{8} \log(2)$$

Example.

$$S(t) = \int_0^1 \frac{\log(1+tx)}{x\sqrt{1-x^2}} dx \quad (t \geq -1)$$

Differentiating under the integral (and assuming $-1 \leq t \leq 1$ for now), we get:

$$S'(t) = \int_0^1 \frac{1}{(1+tx)\sqrt{1-x^2}} dx = \frac{\arccos(t)}{\sqrt{1-t^2}}$$

Integrating yields:

$$S(t) = -\frac{\arccos(t)^2}{2} + C$$

Where C is a constant. Note that the integrand is zero when $t = 0$, so $S(0) = 0$ and so $C = \frac{\pi^2}{8}$. Thus:

$$\int_0^1 \frac{\log(1+tx)}{x\sqrt{1-x^2}} dx = \frac{\pi^2}{8} - \frac{\arccos(t)^2}{2}$$

If $t > 1$, we instead have:

$$S'(t) = \frac{\operatorname{arccosh}(t)}{\sqrt{t^2-1}}$$

This time, integrating yields:

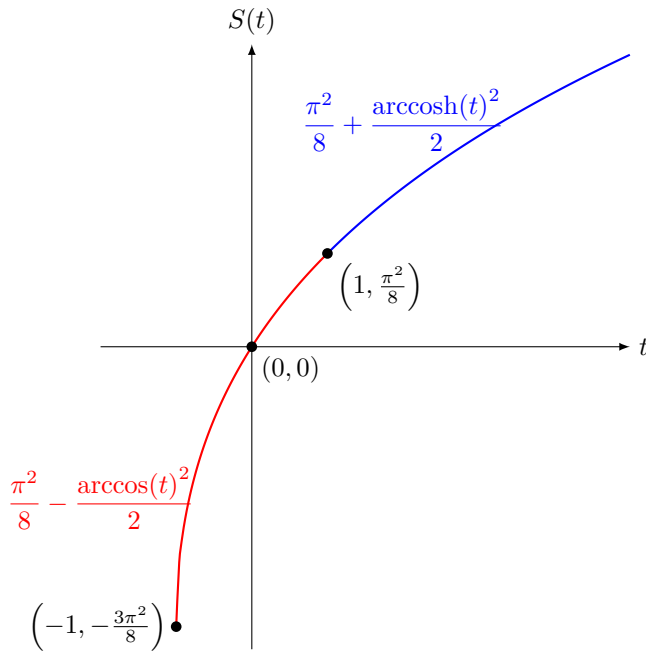
$$S(t) = \frac{\operatorname{arccosh}(t)^2}{2} + K$$

Where K is a constant. As we have already shown above, we have $S(1) = \frac{\pi^2}{8}$ and so $K = \frac{\pi^2}{8}$. Thus:

$$\int_0^1 \frac{\log(1+tx)}{x\sqrt{1-x^2}} dx = \begin{cases} \frac{\pi^2}{8} - \frac{\arccos(t)^2}{2} & -1 \leq t \leq 1 \\ \frac{\pi^2}{8} + \frac{\operatorname{arccosh}(t)^2}{2} & t > 1 \end{cases}$$

Note that the function on the right, although defined piecewise, is analytic at every $t > -1$ (including $t = 1$).

$$\sum_{n=1}^{\infty} \frac{(-2)^{n+1}((n-1)!)^2}{(2n)!} (x-1)^n = \begin{cases} -\arccos(x)^2 & -1 \leq x \leq 1 \\ \operatorname{arccosh}(x)^2 & 1 \leq x \leq 3 \end{cases}$$



Example.

$$S(t) = \int_0^{\frac{\pi}{2}} \frac{\arctan(t \tan(x))}{\tan(x)} dx$$

Differentiating under the integral, we get:

$$S'(t) = \int_0^{\frac{\pi}{2}} \frac{1}{1 + t^2 \tan^2(x)} dx = \frac{\pi}{2(t+1)}$$

Integrating yields:

$$S(t) = \frac{\pi}{2} \log(t+1) + C$$

Where C is a constant. Note that the integrand is zero when $t = 0$, so $S(0) = 0$ and so $C = 0$. Thus:

$$\int_0^{\frac{\pi}{2}} \frac{\arctan(t \tan(x))}{\tan(x)} dx = \frac{\pi}{2} \log(t+1)$$

In particular, setting $t = 1$ yields:

$$\int_0^{\frac{\pi}{2}} \frac{x}{\tan(x)} dx = \frac{\pi}{2} \log(2)$$

Example.

$$S(t) = \int_0^{\infty} e^{-x^2} \cos(tx) dx$$

Differentiating under the integral, we get:

$$S'(t) = \int_0^{\infty} -xe^{-x^2} \sin(tx) dx$$

Integrating by parts yields:

$$S'(t) = \left[-\frac{1}{2} e^{-x^2} \sin(tx) \right]_{x=0}^{x=\infty} - \int_0^{\infty} -\frac{t}{2} e^{-x^2} \cos(tx) dx = \frac{t}{2} \int_0^{\infty} e^{-x^2} \cos(tx) dx$$

Thus $S'(t) = \frac{t}{2} S(t)$. This is a linear first-order ODE in $S(t)$, whose solution is $S(t) = C e^{-\frac{t^2}{4}}$. To find C , note that:

$$S(0) = \int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

Thus:

$$\int_0^{\infty} e^{-x^2} \cos(tx) dx = \frac{\sqrt{\pi}}{2} e^{-\frac{t^2}{4}}$$

Example. This integral was discovered by Laplace in 1810:

$$S(t) = \int_0^{\infty} \frac{\cos(tx)}{x^2 + a^2} dx$$

Differentiating under the integral, we get:

$$S'(t) = \int_0^\infty -\frac{x \sin(tx)}{x^2 + a^2} dx \quad (\text{Convergent for } t \neq 0)$$

Differentiating again, we get:

$$S''(t) = \int_0^\infty -\frac{x^2 \cos(tx)}{x^2 + a^2} dx \quad (\text{Divergent})$$

This is a problem, as the integral on the right does not converge. Instead, we can integrate the expression for $S(t)$ by parts:

$$S(t) = \left[\frac{\sin(tx)}{t(x^2 + a^2)} \right]_{x=0}^{x=\infty} - \int_0^\infty -\frac{2x \sin(tx)}{t(x^2 + a^2)^2} dx = \frac{2}{t} \int_0^\infty \frac{x \sin(tx)}{(x^2 + a^2)^2} dx$$

Call the integral on the right $K(t)$. Now we can differentiate under the integral:

$$\begin{aligned} K'(t) &= \int_0^\infty \frac{x^2 \cos(tx)}{(x^2 + a^2)^2} dx \\ &= \int_0^\infty \frac{\cos(tx)}{x^2 + a^2} dx - a^2 \int_0^\infty \frac{\cos(tx)}{(x^2 + a^2)^2} dx \\ &= S(t) - a^2 \int_0^\infty \frac{\cos(tx)}{(x^2 + a^2)^2} dx \end{aligned}$$

Since $K(t) = \frac{t}{2}S(t)$, we can rewrite the above equation as follows:

$$\begin{aligned} \frac{t}{2}S'(t) + \frac{1}{2}S(t) &= S(t) - a^2 \int_0^\infty \frac{\cos(tx)}{(x^2 + a^2)^2} dx \\ \frac{t}{2}S'(t) - \frac{1}{2}S(t) &= -a^2 \int_0^\infty \frac{\cos(tx)}{(x^2 + a^2)^2} dx \end{aligned}$$

Differentiating under the integral again, we get:

$$\frac{t}{2}S''(t) + \frac{1}{2}S'(t) - \frac{1}{2}S'(t) = a^2 \int_0^\infty \frac{x \sin(tx)}{(x^2 + a^2)^2} dx$$

The integral on the right is simply $K(t) = \frac{t}{2}S(t)$. This yields:

$$\frac{t}{2}S''(t) = a^2 \frac{t}{2}S(t)$$

Thus $S''(t) = a^2 S(t)$. This is a linear second-order ODE in $S(t)$ with constant coefficients, whose solution is $S(t) = C_1 e^{at} + C_2 e^{-at}$. To find C_1 and C_2 , we need two equations. First, we have:

$$S(0) = \int_0^\infty \frac{1}{x^2 + a^2} dx = \frac{\pi}{2a}$$

Thus $C_1 + C_2 = \frac{\pi}{2a}$. Second, we have:

$$\lim_{t \rightarrow \infty} S(t) = \lim_{t \rightarrow \infty} \frac{2}{t} \int_0^\infty \frac{x \sin(tx)}{(x^2 + a^2)^2} dx = 0$$

This can also be seen by applying the [Riemann-Lebesgue lemma](#) directly to $S(t)$. Thus $C_1 = 0$, and so $C_2 = \frac{\pi}{2a}$. This yields:

$$S(t) = \frac{\pi}{2a} e^{-at} \quad (t \geq 0)$$

However, this is only valid for $t \geq 0$, as $S(t)$ is not differentiable at $t = 0$. To handle the case $t < 0$, we can use the fact that $S(t)$ is an even function of t (or alternatively, use the values $S(0) = \frac{\pi}{2a}$ and $\lim_{t \rightarrow -\infty} S(t) = 0$ to solve for C_1 and C_2). Thus, we have:

$$S(t) = \frac{\pi}{2a} e^{-a|t|}$$

Example. Define $f : [0, 1] \times \mathbb{R} \rightarrow \mathbb{R}$ as follows:

$$f(x, t) = \begin{cases} \frac{xt^3}{(x^2 + t^2)^2} & (x, t) \neq (0, 0) \\ 0 & (x, t) = (0, 0) \end{cases}$$

Then for all $t \in \mathbb{R}$, we have $S(t) = \int_0^1 f(x, t) dx = \frac{t}{2(t^2 + 1)}$. Thus $S'(t) = \frac{1-t^2}{2(t^2 + 1)^2}$, and so $S'(0) = \frac{1}{2}$. However,

$$\frac{\partial f}{\partial t} = \begin{cases} \frac{xt^2(3x^2 - t^2)}{(x^2 + t^2)^3} & (x, t) \neq (0, 0) \\ 0 & (x, t) = (0, 0) \end{cases} \quad \therefore \int_0^1 \frac{\partial f}{\partial t}(x, 0) dx = \int_0^1 0 dx = 0$$

The problem here is that $\frac{\partial f}{\partial t}$ is not continuous (as a function of two variables) at $(0, 0)$, as $\frac{\partial f}{\partial t}(t, t) = \frac{1}{4t}$ (for $t \neq 0$), which blows up as $t \rightarrow 0^+$.

1 Other versions

The Leibniz integral rule can be reformulated using the Lebesgue integral:

Theorem 1.1 (Leibniz Integral Rule – Lebesgue Integral). Suppose $(X, \mathcal{A}, \mu)$ is a measure space, $I \subseteq \mathbb{R}$ is an open interval and $f : X \times I \rightarrow \mathbb{R}$ satisfies the following:

1. The function $x \mapsto f(x, t)$ is integrable for all $t \in I$
2. The function $f \mapsto f(x, t)$ is differentiable for all $x \in X$
3. There is an integrable function $g : X \rightarrow \mathbb{R}$ such that $\left| \frac{\partial f}{\partial t}(x, t) \right| \leq g(x)$ for all $x \in X$ and all $t \in I$

Then the function $x \mapsto \frac{\partial f}{\partial t}$ is integrable for all $t \in U$, and:

$$\frac{d}{dt} \left(\int_X f(x, t) d\mu(x) \right) = \int_X \frac{\partial f}{\partial t}(x, t) d\mu(x)$$

Proof. Suppose $(h_n)_{n=1}^\infty$ is a sequence in $\mathbb{R}$ and $\lim_{n \rightarrow \infty} h_n = 0$. For each $n \in \mathbb{N}$, define $g_n : X \rightarrow \mathbb{R}$ as follows:

$$g_n(x) = \frac{f(x, t + h_n) - f(x, t)}{h_n} - \frac{\partial f}{\partial t}(x, t)$$

Then $\lim_{n \rightarrow \infty} g_n(x) = 0$ for all $x \in X$. By [Lagrange's mean value theorem](#), we have:

$$\frac{f(x, t + h_n) - f(x, t)}{h_n} = \frac{\partial f}{\partial t} \text{ for some } \tau_n \text{ between } t \text{ and } t + h_n$$

Thus for all $n \in \mathbb{N}$ and all $x \in X$, we have:

$$|g_n(x)| = \left| \frac{\partial f}{\partial t} - \frac{\partial f}{\partial t}(x, t) \right| \leq \left| \frac{\partial f}{\partial t} \right| + \frac{\partial f}{\partial t}(x, t) \leq g(x) + g(x) = 2g(x)$$

By the [dominated convergence theorem](#), we have:

$$\frac{1}{h_n} \int_X (f(x, t + h_n) - f(x, t)) d\mu(x) - \int_X \frac{\partial f}{\partial t}(x, t) d\mu(x) = \int_X g_n(x) d\mu(x) \rightarrow 0 \text{ as } n \rightarrow \infty$$

As $n \rightarrow \infty$, the first integral tends to $\frac{d}{dt} \left(\int_X f(x, t) d\mu(x) \right)$. Since this holds for every sequence $(h_n)_{n=1}^\infty$ that converges to zero, we have:

$$\frac{d}{dt} \left(\int_X f(x, t) d\mu(x) \right) = \int_X \frac{\partial f}{\partial t}(x, t) d\mu(x) \quad \text{😎}$$

The Leibniz integral rule can also be adapted to the case where the limits are not constants but functions of t :

Theorem 1.2 (Leibniz Integral Rule – Variable Limits). Suppose $I \subseteq \mathbb{R}$ is an open interval, $a, b : I \rightarrow \mathbb{R}$ are differentiable on I such that $a(t), b(t) \in [A, B]$ for all $t \in I$, and $f : [a, b] \times I \rightarrow \mathbb{R}$, $(x, t) \mapsto f(x, t)$ such that f and $\frac{\partial f}{\partial t}$ are continuous on $[A, B] \times I$. Define $S : I \rightarrow \mathbb{R}$ as follows:

$$S(t) = \int_{a(t)}^{b(t)} f(x, t) dx$$

Then S is differentiable on I and:

$$S'(t) = b'(t)f(b(t), t) - a'(t)f(a(t), t) + \int_{a(t)}^{b(t)} \frac{\partial f}{\partial t}(x, t) dx$$

Proof. Define $G : [A, B] \times [A, B] \times I \rightarrow \mathbb{R}$ as follows:

$$G(a, b, t) = \int_a^b f(x, t) dx$$

In other words, G is a function of three independent variables a, b, t . By the [fundamental theorem of calculus](#), we have:

$$\frac{\partial G(a, b, t)}{\partial b} = f(b, t) \qquad \frac{\partial G(a, b, t)}{\partial a} = -f(a, t)$$

By [Theorem 0.1](#), we also have:

$$\frac{\partial G(a, b, t)}{\partial t} = \int_a^b \frac{\partial f}{\partial t}(x, t) dx$$

Thus, by the multivariable chain rule, we have:

$$\begin{aligned}\frac{d}{dt}G(a(t), b(t), t) &= \frac{db(t)}{dt} \frac{\partial G}{\partial b}(a(t), b(t), t) + \frac{da(t)}{dt} \frac{\partial G}{\partial a}(a(t), b(t), t) + \frac{dt}{dt} \frac{\partial G}{\partial t}(a(t), b(t), t) \\ &= b'(t) \frac{\partial G}{\partial b}(a(t), b(t), t) + a'(t) \frac{\partial G}{\partial a}(a(t), b(t), t) + \frac{\partial G}{\partial t}(a(t), b(t), t) \\ &= b'(t)f(b(t), t) - a'(t)f(a(t), t) + \int_{a(t)}^{b(t)} \frac{\partial f}{\partial t}(x, t) dx\end{aligned}$$



Fix $t_0 \in I$.

$$\begin{aligned}S(t) &= \int_{a(t)}^{b(t)} f(x, t) dx \\ &= \int_{b(t_0)}^{b(t)} f(x, t) dx - \int_{a(t_0)}^{a(t)} f(x, t) dx + \int_{a(t_0)}^{b(t_0)} f(x, t) dx \\ &= \int_{b(t_0)}^{b(t)} f(x, t_0) dx - \int_{a(t_0)}^{a(t)} f(x, t_0) dx + \int_{a(t_0)}^{b(t_0)} f(x, t) dx \\ &\quad + \int_{b(t_0)}^{b(t)} (f(x, t) - f(x, t_0)) dx - \int_{a(t_0)}^{a(t)} (f(x, t) - f(x, t_0)) dx\end{aligned}$$

We first show that $S'_1(t) = b'(t)f(b(t), t)$. By the **fundamental theorem of calculus** and the **chain rule**, we have:

$$S'_1(t) = b'(t)f(b(t), t) \qquad S'_2(t) = a'(t)f(a(t), t)$$

By **Theorem 0.1**, we also have:

$$S'_3(t) = \int_{a(t_0)}^{b(t_0)} \frac{\partial f}{\partial t}(x, t) dx$$

Since b is differentiable at t_0 , it is continuous at t_0 , so $\lim_{t \rightarrow t_0} |b(t) - b(t_0)| = 0$. Since $\frac{\partial f}{\partial t}$ is continuous at t_0 , it is bounded in a neighborhood of t_0 , i.e. there exists

$$\begin{aligned}\frac{S_4(t) - S_4(t_0)}{t - t_0} &= \frac{1}{t - t_0} \int_{b(t_0)}^{b(t)} (f(x, t) - f(x, t_0)) dx \\ &= \int_{b(t_0)}^{b(t)} \frac{f(x, t) - f(x, t_0)}{t - t_0} dx \\ &\leq |b(t) - b(t_0)| \sup_{\xi \in B} \int_{b(t_0)}^{b(t)} \frac{f(x, t) - f(x, t_0)}{t - t_0} dx\end{aligned}$$

Example (Variant of Serret's integral).

$$S(t) = \int_0^t \frac{\log(1 + tx)}{1 + x^2} dx$$

Differentiating under the integral, we get:

$$\begin{aligned}S'(t) &= \frac{\log(1 + t^2)}{1 + t^2} - 0 + \int_0^t \frac{x}{(1 + tx)(1 + x^2)} dx \\ &= \frac{\log(1 + t^2)}{1 + t^2} - 0 - \frac{\log(1 + t^2)}{1 + t^2} + \frac{t \arctan(t)}{1 + t^2} + \frac{\log(1 + t^2)}{2(1 + t^2)} \\ &= \frac{t \arctan(t)}{1 + t^2} + \frac{\log(1 + t^2)}{2(1 + t^2)}\end{aligned}$$

Integrating the first term by parts yields:

$$\int \frac{t \arctan(t)}{1 + t^2} dt = \frac{\arctan(t) \log(1 + t^2)}{2} - \int \frac{\log(1 + t^2)}{2(1 + t^2)} dt$$

Thus the second term cancels out, and so:

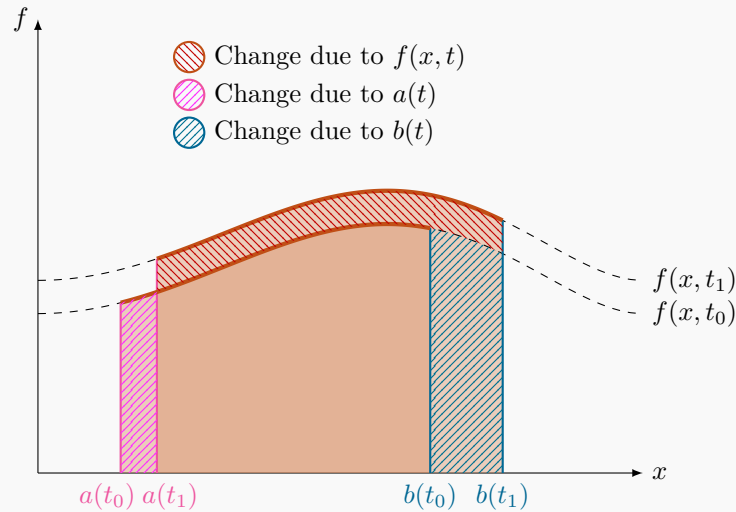
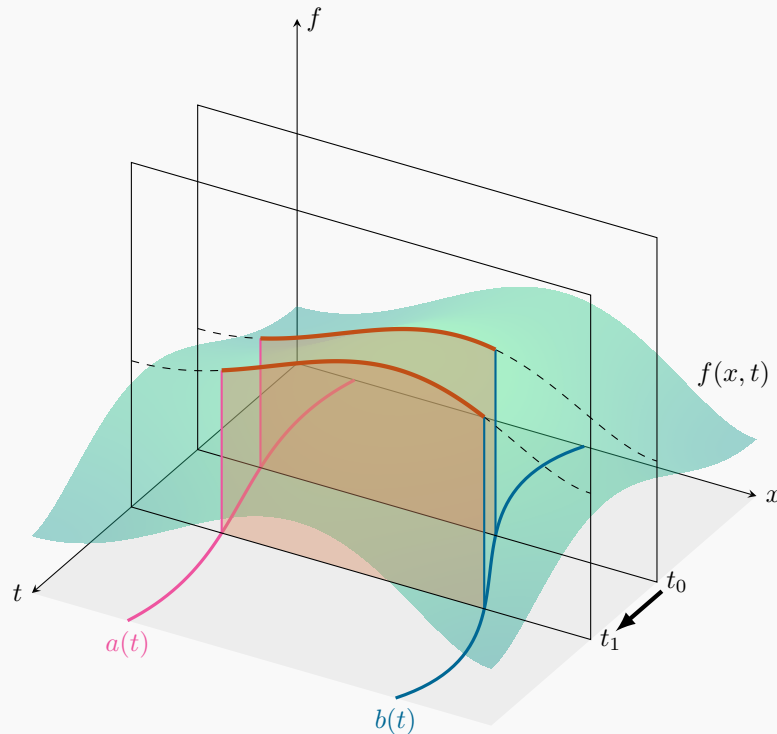
$$S(t) = \frac{\arctan(t) \log(1 + t^2)}{2} + C$$

Where C is a constant. As with Serret's integral, the integrand vanishes when $t = 0$, so $S(0) = 0$ and so $C = 0$. Thus:

$$\int_0^t \frac{\log(1 + tx)}{1 + x^2} dx = \frac{\arctan(t) \log(1 + t^2)}{2}$$

The Leibniz Integral Rule (Variable Limits)

$$\frac{d}{dt} \left(\int_{a(t)}^{b(t)} f(x, t) dx \right) = b'(t)f(b(t), t) - a'(t)f(a(t), t) + \int_{a(t)}^{b(t)} \frac{\partial f}{\partial t}(x, t) dx$$



We now use the Leibniz integral rule to prove the **fundamental theorem of algebra**:

Theorem 1.3 (Fundamental Theorem of Algebra). Every polynomial $p : \mathbb{C} \rightarrow \mathbb{C}$ has at least one root.

Proof. Suppose $p(z) \neq 0$ for all $z \in \mathbb{C}$. Define $I : [0, \infty) \rightarrow \mathbb{C}$ as follows:

$$I(r) = \int_0^{2\pi} \frac{1}{p(re^{i\theta})} d\theta$$

This is well-defined as $p(z)$ is never zero, so $\frac{1}{p(z)}$ is continuous on $\mathbb{C}$.

We first show that $\lim_{r \rightarrow 0^+} I(r) = I(0)$, i.e. I is continuous at 0. Since $\frac{1}{p(z)}$ is continuous at 0, for any $\varepsilon > 0$, there

exists $\delta > 0$ such that $\left| \frac{1}{p(z)} - \frac{1}{p(0)} \right| < \frac{\varepsilon}{2\pi}$ for all $z \in \mathbb{C}$, $|z| < \delta$. Thus for all $0 \leq r < \delta$, we have:

$$|I(r) - I(0)| = \left| \int_0^{2\pi} \left(\frac{1}{p(re^{i\theta})} - \frac{1}{p(0)} \right) d\theta \right| \leq \int_0^{2\pi} \left| \frac{1}{p(re^{i\theta})} - \frac{1}{p(0)} \right| d\theta < \int_0^{2\pi} \frac{\varepsilon}{2\pi} d\theta = \varepsilon$$

Thus $\lim_{r \rightarrow 0^+} I(r) = I(0)$.

We now show that $\lim_{r \rightarrow \infty} I(r) = 0$. Suppose the leading (highest-degree) term of $p(z)$ is az^d . Then we have $\lim_{r \rightarrow \infty} \frac{|p(re^{i\theta})|}{|re^{i\theta}|^d} = |a|$, so there exists $R > 0$ such that $|p(z)| > \frac{|a|}{2} r^d$ for all $r > R$. Thus for all $r > R$, we have:

$$|I(r)| \leq \int_0^{2\pi} \frac{1}{|p(re^{i\theta})|} d\theta \leq \int_0^{2\pi} \frac{2}{|a|r^d} d\theta = \frac{4\pi}{|a|r^d}$$

Since the right side tends to zero as $r \rightarrow \infty$, we have $\lim_{r \rightarrow \infty} I(r) = 0$.

Note that:

$$\frac{\partial}{\partial \theta} \left(\frac{1}{p(re^{i\theta})} \right) = -ire^{i\theta} \frac{p'(re^{i\theta})}{p(re^{i\theta})^2} = ir \frac{\partial}{\partial r} \left(\frac{1}{p(re^{i\theta})} \right)$$

Thus, differentiating under the integral, we get:

$$I'(r) = \int_0^{2\pi} \frac{\partial}{\partial r} \left(\frac{1}{p(re^{i\theta})} \right) d\theta = \int_0^{2\pi} \frac{1}{ir} \frac{\partial}{\partial \theta} \left(\frac{1}{p(re^{i\theta})} \right) d\theta = \frac{1}{ir} \left(\frac{1}{p(re^{i\theta})} \right)_{\theta=0}^{\theta=2\pi} = \frac{1}{ir} \left(\frac{1}{p(r)} - \frac{1}{p(r)} \right) = 0$$

Thus I is constant on $[0, \infty)$. This is a contradiction, as $\lim_{r \rightarrow \infty} I(r) = 0$ but $\lim_{r \rightarrow 0^+} I(r) = \frac{2\pi}{p(0)} \neq 0$. Thus p must have at least one root, i.e. $p(z) = 0$ for some $z \in \mathbb{C}$. 🤔